ANUP BAJGAIN | Aerospace Engineer

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EDUCATION

Institute of Engineering, Pulchowk Campus, Tribhuvan University — Lalitpur, Nepal
Bachelor of Engineering in Aerospace Engineering — 2022–2026

- Cumulative Percentage: 84.73%
- Academic Standing: Top of class overall
- Relevant Coursework: Aerodynamics, Flight Dynamics, Propulsion Systems, Avionics, Control Systems, CFD, Aircraft Manufacturing Process, Aircraft Maintenance Process

EXPERIENCE

Yeti Airlines — Kathmandu, Nepal
Operations Engineering Intern — 2025–2026 (6 months)

- Supported OCC and Dispatch operations, assisting with real-time flight monitoring and operational decision-making for the ATR 72-500 fleet
- Assisted with Weight & Balance calculations, pre-flight documentation, and takeoff/landing performance analysis under varying operational conditions
- Maintained operational records and technical documentation in compliance with CAAN regulatory requirements

DBF Pulchowk (AIAA Design/Build/Fly) — IOE Pulchowk
Design Team Member — 2023–2024

- Contributed to development of competition aircraft *Junkiri*
- Collaborated on aerodynamic design, structural optimization, and flight testing
- Contributed to the preparation of technical documentation for competition submission

PROJECTS

Study of the Surface Heating Effect on a Compressible Wall Jet using Rainbow Schlieren Deflectometry
Final Year Project — 2026

- Conducted experimental flow visualization using Rainbow Schlieren Deflectometry (RSD) to measure density gradient fields in a compressible wall-jet configuration
- Performed 3D RANS CFD simulations in ANSYS Fluent and validated results against experimental data
- Developed Python post-processing scripts for quantitative comparison of CFD and experimental density gradient profiles across multiple streamwise stations

Agricultural UAV Design

- Modeled the aircraft geometry in Plane Maker and performed aerodynamic validation using X-Plane flight simulator
- Extracted and analyzed simulated flight data to verify aerodynamic performance metrics against analytical estimates

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Matplotlib, OOP, DSA)

Analysis & Simulation: ANSYS Fluent, OpenFOAM, XFLR5

CAD: CATIA V5, SolidWorks

Others: LaTeX, MS Office

TRAININGS & CERTIFICATIONS

2025: Samsung Innovation Campus – Coding and Programming Certification
2024: Coursera – Machine Learning Specialization
2023: Network Academy – Python Essentials 2
2023: Network Academy – Networking Basics

INTERESTS

Music, Travel, Chess, Football